NETWORKWALKS • CYBERSECURITY INTERNSHIP • BATCH B084

PENETRATION TESTING REPORT

Footprinting & Network Scanning Phases

W2-PM-FINAL | Week 2 Project Work | networkwalks.com + own LAN

Pentester (Cybersecurity Intern)	Junaideen Risny Suha
Program / Batch	Cybersecurity Internship 2026 — B084 (Networkwalks)
Report date	13 November 2026
Modules completed	W2-PM1 — Footprinting with multiple tools (6 tools) W2-PM5 — Network scanning with Zenmap W2-PM-FINAL — This report
Client / Targets	1. networkwalks.com — in scope under the Networkwalks internship authorization 2. My own local area network (LAN)
Permission secured?	**Yes — scope per Networkwalks Letter of Authorization (Appendix A)**
Phases covered	Phase 1: Reconnaissance & Footprinting ✔ Phase 2: Scanning & Network Discovery ✔ Phases 3–5: Upcoming weeks
Classification	Educational — for Networkwalks internship assessment only

Executive summary

Six reconnaissance tools built a complete public profile of networkwalks.com without sending a single attack: a WordPress 7.1.4 site with the WP Download Manager 3.3.76 plugin, served by Apache on HostGator shared hosting (192.232.216.135), protected by a ModSecurity WAF, with mail on the same server and DNSSEC not enabled.

The biggest takeaways are version disclosure, a missing set of browser security headers (no HSTS, CSP or X-Frame-Options), a soft SPF policy and unsigned DNS. None of these is an exploited vulnerability; they are hardening opportunities.

A Zenmap ping scan of my own LAN (192.168.8.0/24) found 3 live hosts in 5.68 seconds: the Huawei home router (192.168.8.1), a device using a randomized private MAC (192.168.8.101) and my own PC (192.168.8.102). I recorded each IP and MAC address and saved the topology as a PDF (Section 6).

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1. Liability Disclaimer

I performed these activities only on systems where written permission was provided (networkwalks.com, under the Networkwalks internship Letter of Authorization) and on devices and a network that I am authorized to use myself. No exploitation, brute-force, denial-of-service or data access was attempted.

All material in this report is for education and research purposes only. Do not use anything here to break the law. The instructor, the authors and Networkwalks are not responsible for any misuse of this knowledge. Unauthorized access is a crime in most countries, even when nothing is damaged.

2. Scope & Rules of Engagement

Item	In scope	Out of scope
Targets	networkwalks.com and its public DNS records; my own LAN subnet	Any other domain, IP address, network or third-party system
Activities	Passive / light footprinting (whois, whatweb, nslookup, curl, wafw00f, dnsrecon); ping-scan host discovery with Zenmap	Exploitation, password attacks, DoS, vulnerability exploitation, social engineering
Data handling	Findings kept for this internship report only	Copying, modifying or deleting any client data

Testing window: 13 November 2026 (Week 2 project period, submission deadline 13 Nov 2026, 23:00).

3. Introduction

Every real attack, and every authorized penetration test, starts with the same question: what can I learn about the target before I touch it? Week 2 answers that question in two phases.

- **Phase 1 — Footprinting (W2-PM1):** I footprinted networkwalks.com with six tools. Each one reveals a different layer: registration, technology stack, IP address, HTTP headers, firewall and DNS.
- **Phase 2 — Scanning (W2-PM5):** I used Zenmap (the Nmap GUI) on Windows to discover every live device on my own LAN, with IP and MAC addresses and a topology map.

For every step I record the exact command, what I observed, a screenshot as evidence, and why the finding matters from an attacker's and a defender's point of view.

4. Lab Environment & Tools

Tool	Purpose	Where it ran
whois	Domain registration details: registrar, dates, name servers	Linux terminal (Debian 12, Google Cloud Shell)
whatweb	Fingerprint web technologies: server, CMS, plugins, IP	Linux terminal
nslookup	Resolve the domain to its IP address	Linux terminal

Tool	Purpose	Where it ran
curl -I	Read the raw HTTP response headers	Linux terminal
wafw00f v2.4.2	Detect a Web Application Firewall	Linux terminal
dnsrecon	Enumerate DNS records: SOA, NS, MX, A, TXT, SPF, SRV	Linux terminal
Zenmap / Nmap 7.993	Host discovery on my LAN: live hosts, IPs, MACs, topology	Windows PC (local LAN)
nmap --iflist	List my PC's interfaces: IP, subnet mask, MAC and default gateway	Zenmap command field (Windows PC)

Why a Linux cloud terminal instead of my Kali VM?

My laptop with the Week 1 Kali Linux VM was away for repair this week. To keep to the deadline without skipping any tool, I installed the exact same six tools (whois, whatweb, dnsutils, wafw00f, dnsrecon) on a Debian-based Linux terminal (Google Cloud Shell) and ran the identical commands from the module guide.

The commands, tools and outputs are the same as on Kali. The only visible difference is the DNS resolver address in the nslookup output (169.254.169.254, the cloud's internal resolver).

5. Phase 1 — Footprinting networkwalks.com (W2-PM1)

Each tool below answers one question about the target. Together they build the full profile summarized in Section 7.

5.1 Task 1: WHOIS — domain registration

```
$ whois networkwalks.com
```

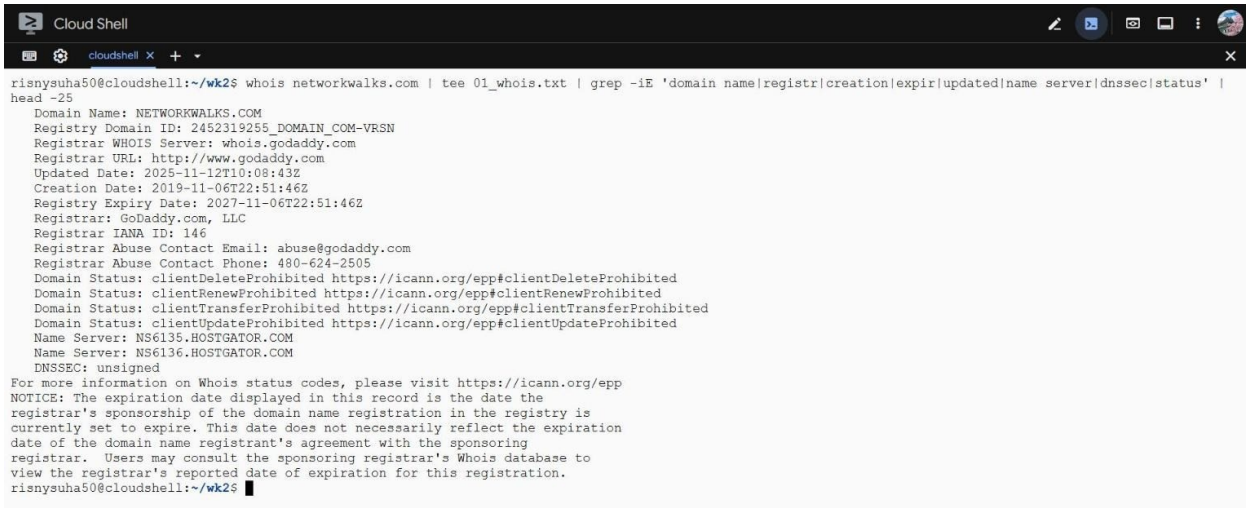


Figure 1: WHOIS output for networkwalks.com

Key findings

Registrar	GoDaddy.com, LLC (IANA ID 146)
Created	06 Nov 2019

Last updated	12 Nov 2025
Expires	06 Nov 2027
Name servers	NS6135.HOSTGATOR.COM, NS6136.HOSTGATOR.COM
Domain status	client Delete / Renew / Transfer / Update Prohibited (registrar lock ON)
DNSSEC	unsigned

How an attacker uses this / what a defender learns

The name servers reveal the hosting provider (HostGator) even though the domain is registered at GoDaddy. Registration dates and the abuse contact are useful for social-engineering pretexts. The good news: all four registrar locks are on, so domain hijacking by transfer is hard.

5.2 Task 2: WhatWeb — technology fingerprint

```
$ whatweb networkwalks.com
```

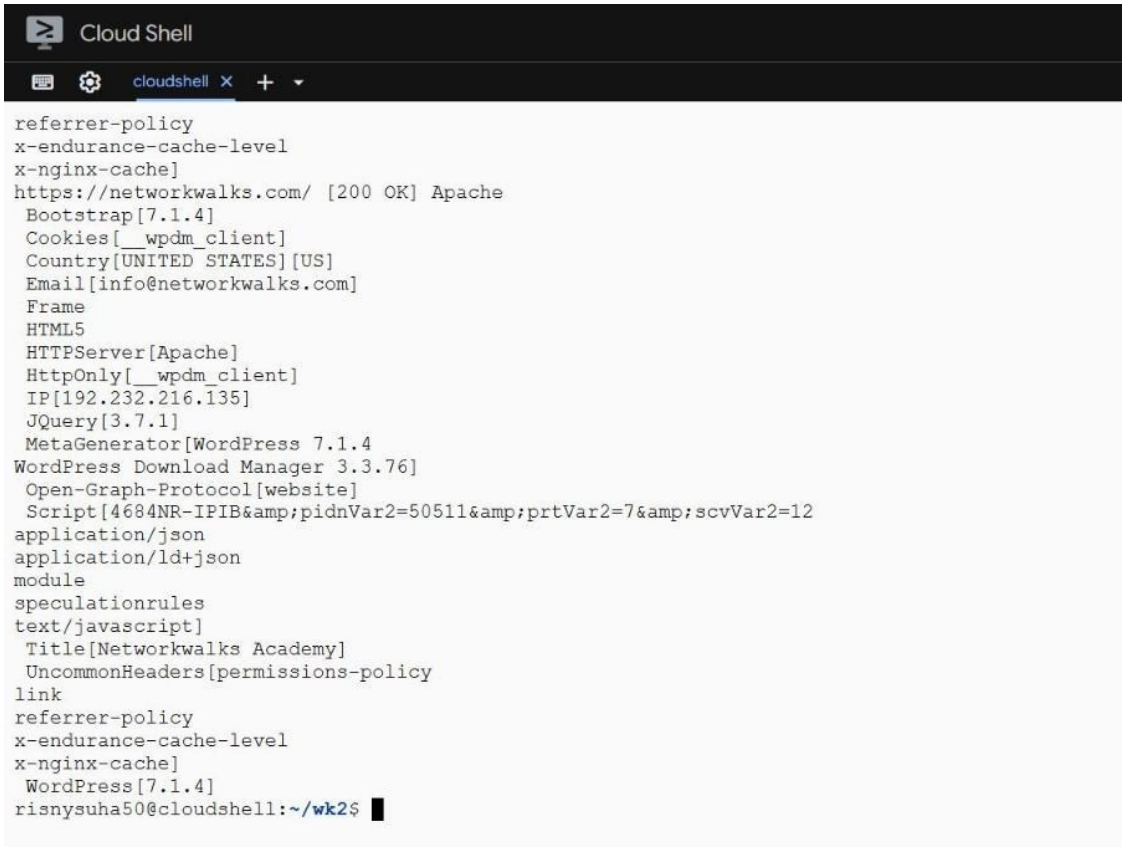


Figure 2: WhatWeb output for networkwalks.com

Key findings

Web server	Apache (HTTP 301 → HTTPS, then 200 OK)
CMS	WordPress 7.1.4
Plugin	WordPress Download Manager 3.3.76

Libraries	jQuery 3.7.1, Bootstrap, HTML5, Open Graph
IP / Country	192.232.216.135 / United States
Exposed email	info@networkwalks.com
Cookie	__wpdm_client (HttpOnly)
Site title	Networkwalks Academy

> **How an attacker uses this / what a defender learns**
> Exact software versions are the attacker's shopping list: they are matched against CVE and WPScan databases to find known exploits. The exposed email is a ready-made phishing target.

5.3 Task 3: Nslookup — DNS resolution

```
$ nslookup networkwalks.com
```



Figure 3: Nslookup output for networkwalks.com

Key findings

Resolved IP (A record)	192.232.216.135
Answer type	Non-authoritative (from a caching resolver)

> **How an attacker uses this / what a defender learns**
> The IP address lets an attacker scan the server directly, look up other websites on the same shared host (reverse-IP), and map the hosting network.

5.4 Task 4: cURL -I — HTTP response headers

```
$ curl -sI https://networkwalks.com
```



Figure 4: cURL -I output for networkwalks.com

Key findings

Status	HTTP/1.1 200 OK
Server banner	Apache
Hidden endpoints	/wp-json/ (WordPress REST API), /wp-json/wp/v2/pages/53
Caching stack	X-nginx-cache: WordPress; X-Endurance-Cache-Level: 0 (Endurance / HostGator)
Cookie flags	secure; HttpOnly ✔️
Referrer-Policy	no-referrer-when-downgrade
Missing headers	Strict-Transport-Security, Content-Security-Policy, X-Frame-Options, X-Content-Type-Options

> **How an attacker uses this / what a defender learns**
>
> Headers leak the stack without loading the page. The open /wp-json/ API is a common starting point for WordPress user enumeration. The missing security headers mean the browser gets no HSTS, clickjacking or content-type protection from the site.

5.5 Task 5: Wafw00f — WAF detection

```
$ wafw00f https://networkwalks.com
```

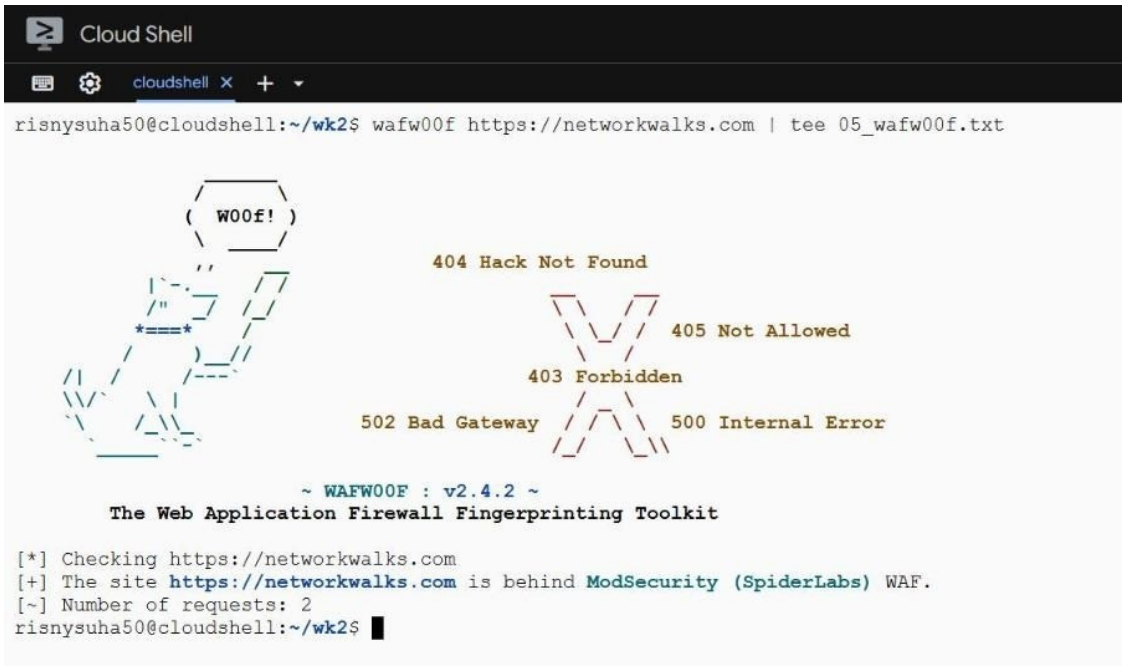


Figure 5: Wafw00f output for networkwalks.com

Key findings

WAF detected	ModSecurity (SpiderLabs)
Requests needed	2

> **How an attacker uses this / what a defender learns**
>
> Knowing a WAF is present changes the whole attack plan: noisy or naive payloads will be blocked and logged, so an attacker must slow down or attempt bypass techniques. For the defender, this is a strong positive control.

5.6 Task 6: DNSRecon — DNS enumeration

```
$ dnsrecon -d networkwalks.com
```

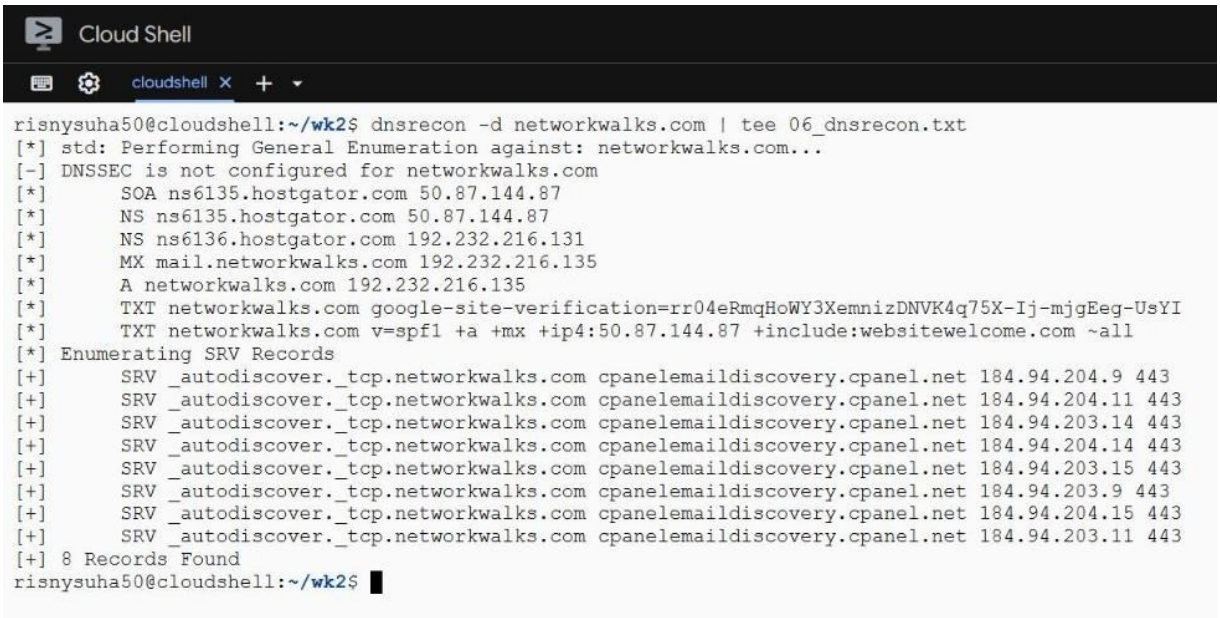


Figure 6: DNSRecon output for networkwalks.com

Key findings

DNSSEC	Not configured
SOA / NS	ns6135.hostgator.com (50.87.144.87), ns6136.hostgator.com (192.232.216.131)
MX (mail)	mail.networkwalks.com → 192.232.216.135 (same server as the website)
A	networkwalks.com → 192.232.216.135
SPF (TXT)	v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
Other TXT	google-site-verification token
SRV	8 × _autodiscover._tcp → cpanelemaildiscovery.cpanel.net (cPanel)

> **How an attacker uses this / what a defender learns**
>
> DNS maps the whole infrastructure: web and mail share one IP (one compromise affects both), the SRV records confirm cPanel hosting, and the SPF policy ends in ~all (soft-fail), which is weaker than -all against email spoofing. Without DNSSEC, DNS answers can't be cryptographically verified.

5.7 What changed since the instructor's guide

Running the tools myself on 13 Nov 2026 showed that the target had been updated since the module guide was written. This proves the results are live, and shows why footprinting must be repeated before every engagement:

Item	In the module guide	My scan (13 Nov 2026)
WordPress	7.0.4	7.1.4 (patched / upgraded)
WP Download Manager	3.3.58	3.3.76 (patched / upgraded)
DNS server software (dnsrecon)	Bind 9.16.23 version shown	Version no longer disclosed

Item	In the module guide	My scan (13 Nov 2026)
IP address / WAF	192.232.216.135 / ModSecurity	Unchanged

6. Phase 2 — Network Scanning with Zenmap (W2-PM5)

Zenmap is the official graphical interface for Nmap. I installed it on my Windows PC and used a Ping Scan (host discovery, no port scan) on my own LAN to answer the seven module tasks.

6.1 Task 1: Download & install Zenmap

I downloaded the official Windows self-installer nmap-7.993-setup.exe from nmap.org/download.html and installed Nmap, Zenmap and the Npcap packet-capture driver with the default options.

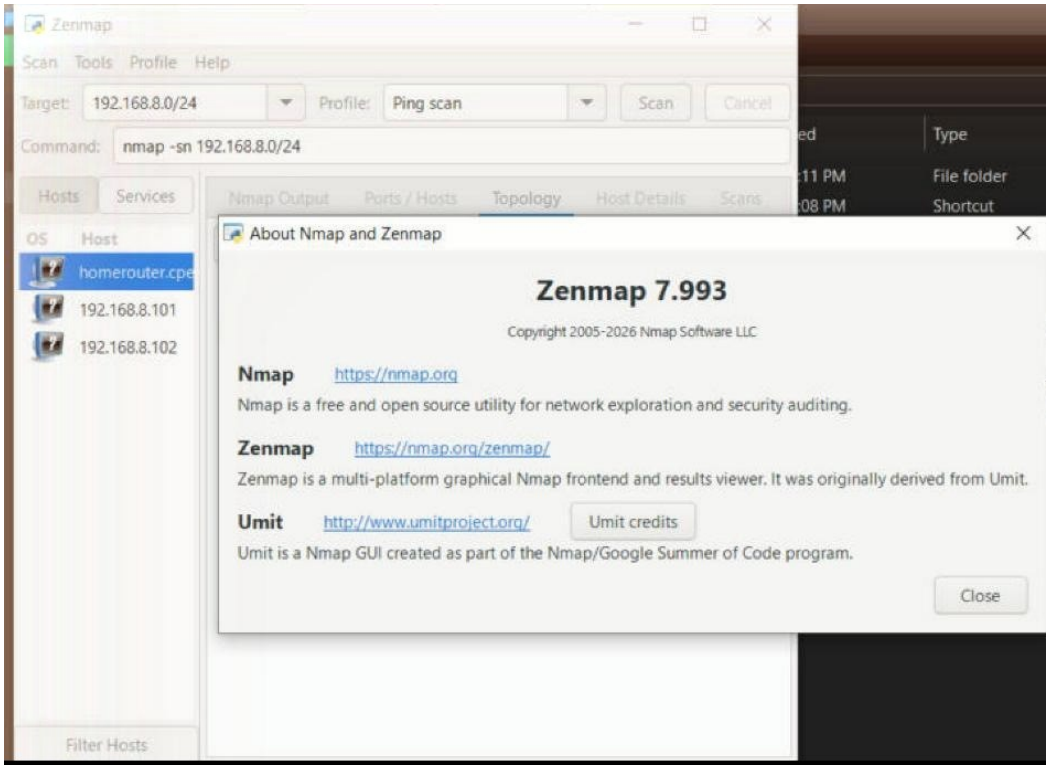


Figure 7: Zenmap 7.993 installed and running (Help → About)

6.2 Task 2: Find my local IP address & LAN subnet

The guide uses ipconfig in CMD. I got the same information inside Zenmap itself with nmap --iflist, which lists every network interface (IP/mask, MAC) and the routing table (default gateway). The active adapter is eth3, the only interface that is up with a 192.168.x.x address.

```
$ nmap --iflist
```

Local IP address	192.168.8.102
Subnet mask	255.255.255.0
LAN subnet (CIDR)	192.168.8.0/24
Default gateway	192.168.8.1 (homerouter.cpe)

My MAC address	D4:6D:6D:72:CF:27

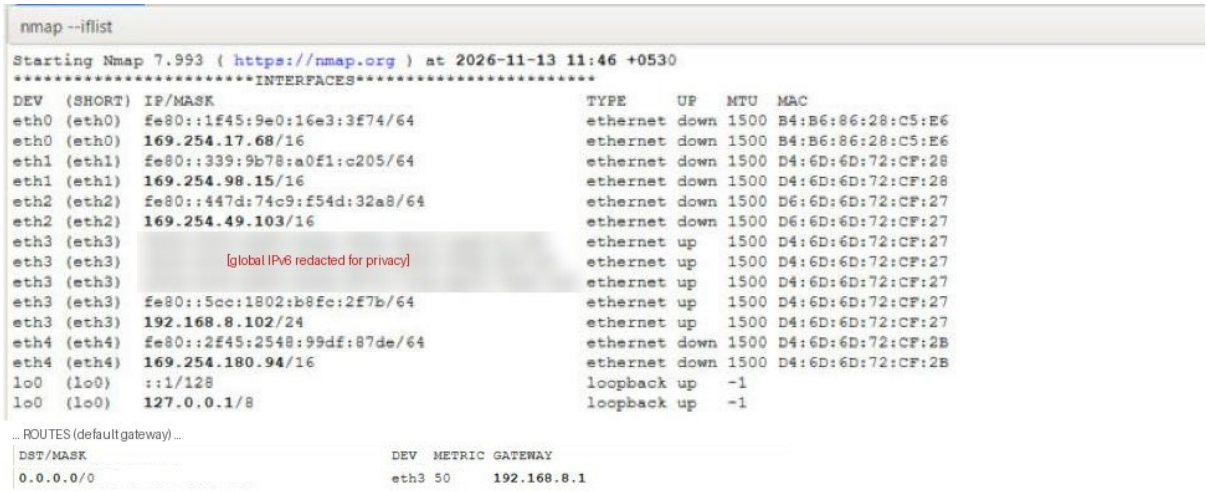


Figure 8: nmap --iflist: my IP 192.168.8.102/24, MAC D4:6D:6D:72:CF:27, gateway 192.168.8.1 (global IPv6 redacted)

6.3 Task 3: Find the live hosts in my subnet (Ping Scan)

```
$ nmap -sn 192.168.8.0/24
Zenmap → Target: my subnet → Profile: Ping scan → Scan
```

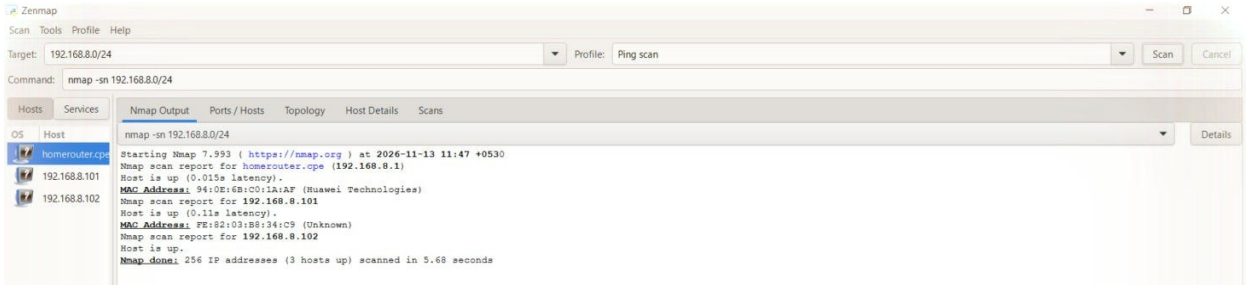


Figure 9: Zenmap Ping Scan of my LAN subnet

6.4 Tasks 4–6: How many live hosts, their IP and MAC addresses

Answer (Task 4): 3 hosts are live in 192.168.8.0/24 (including my own PC).

#	IP address	MAC address	Vendor / role
1	192.168.8.1	94:0E:6B:C0:1A:AF	Huawei Technologies — home router / default gateway (homerouter.cpe)
2	192.168.8.101	FE:82:03:B8:34:C9	Unknown vendor — randomized (private) MAC, typical of a phone using Wi-Fi privacy
3	192.168.8.102	D4:6D:6D:72:CF:27	My Windows PC — the scanner (MAC from nmap --iflist)

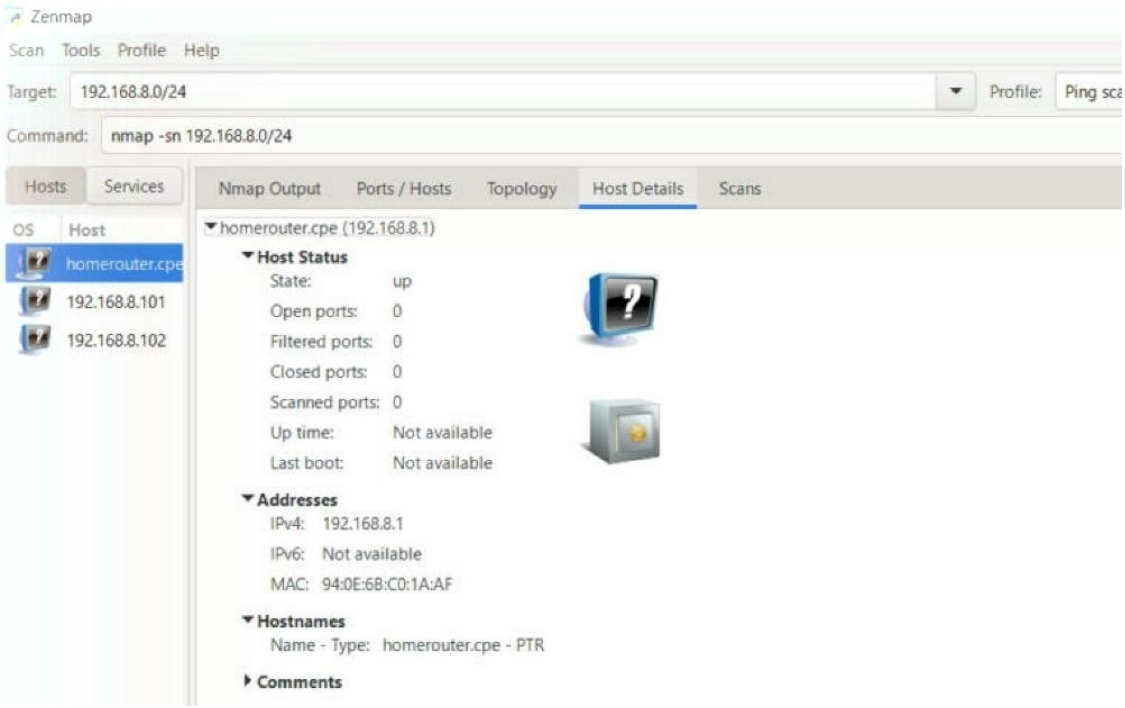


Figure 10: Host Details: the gateway 192.168.8.1 with its Huawei MAC address

Note: Nmap can't read the MAC address of the scanning PC itself over the network (its own row shows no MAC in the scan), so my own MAC comes from nmap --iflist, the same idea as the guide's ipconfig /all tip.

6.5 Task 7: Topology saved as PDF

In the Topology tab I turned on the legend and read it: every host is a green circle ("fewer than 3 open ports", which here means not port-scanned because this was a ping scan), and the dashed lines mean no traceroute information, because a ping scan doesn't trace routes. All three hosts sit on the first ring around localhost, so they are one hop away on the same LAN. Zenmap always draws the scanning machine as "localhost" at the centre, so my PC appears twice: as localhost and as 192.168.8.102. I then used Save Graphic → PDF to save it to the desktop as Zenmap_Topology_Risny_WK2.pdf (included in the evidence folder).

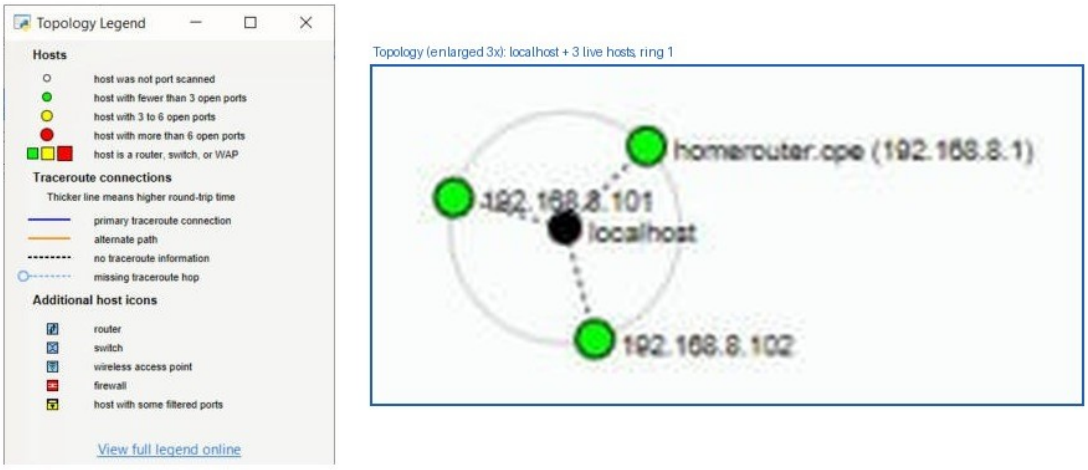


Figure 11: Zenmap topology with the legend: localhost (the scanner) at the centre and the 3 live hosts one hop away

Scan observations

Scan result: 256 IP addresses scanned, 3 hosts up, in 5.68 seconds (ping scan only, no ports probed).

192.168.8.101 uses a locally administered MAC (the second-lowest bit of the first byte, FE, is set). Phones and laptops randomize their Wi-Fi MAC for privacy, so Nmap can't name the vendor. In a real assessment this device

> would need to be identified by its owner rather than trusted by default.
>
> The gateway answers to the hostname homerouter.cpe and has a Huawei MAC prefix, which identifies the make of the router without logging in to it.

7. Consolidated Target Profile

This is the picture an attacker would hold after Phase 1, built only from public information:

Organization / site	Networkwalks Academy — networkwalks.com
Registrar / host	GoDaddy (registrar) • HostGator / Endurance shared hosting with cPanel
IP addresses	Web + mail: 192.232.216.135 • DNS: 50.87.144.87, 192.232.216.131
Web stack	Apache + nginx cache • WordPress 7.1.4 • WP Download Manager 3.3.76 • jQuery 3.7.1
Security controls	ModSecurity WAF ✔️ • HTTPS redirect ✔️ • registrar lock ✔️ • secure/HttpOnly cookie ✔️
Gaps	DNSSEC off • SPF ~all • no HSTS / CSP / X-Frame-Options / X-Content-Type-Options • version banners public
Contact points	info@networkwalks.com (site) • abuse@godaddy.com (registrar)

8. Risk Analysis

The ratings below are for observations from passive reconnaissance and host discovery. They are not confirmed vulnerabilities; confirming any of them would need further authorized testing.

#	Finding	Evidence	Potential impact	Risk
1	CMS & plugin versions disclosed	whatweb: WordPress 7.1.4, WPDM 3.3.76 in meta generator	Speeds up matching to known CVEs	Medium
2	Browser security headers missing	curl -I: no HSTS, CSP, X-Frame-Options, X-Content-Type-Options	SSL-strip, clickjacking, XSS impact not reduced	Medium
3	DNSSEC not enabled	whois + dnsrecon: unsigned	DNS spoofing / cache poisoning can't be detected	Medium
4	Soft SPF policy	SPF ends in ~all (soft-fail)	Spoofed email may still be delivered	Medium
5	Web and mail on one host	MX and A both → 192.232.216.135	One compromise affects website and email	Low
6	WordPress REST API exposed	Link header: /wp-json/	Possible user enumeration	Low
7	Public contact email	whatweb: info@networkwalks.com	Phishing / social-engineering target	Low
8	WAF in place	wafw00f: ModSecurity (SpiderLabs)	Positive control — blocks naive attacks	Info

#	Finding	Evidence	Potential impact	Risk
9	Devices visible on my LAN	Zenmap: 3 live hosts with MACs	Unknown devices could be rogue or unpatched	Low

Risk key: High • Medium • Low • Info (positive control)

9. Recommendations

#	Recommendation	How
1	Hide version banners	Remove the WordPress generator meta tag and plugin version strings; set Apache ServerTokens Prod.
2	Add security headers	Strict-Transport-Security (HSTS), Content-Security-Policy, X-Frame-Options: SAMEORIGIN, X-Content-Type-Options: nosniff, Referrer-Policy: strict-origin-when-cross-origin.
3	Enable DNSSEC	Turn on DNSSEC at the DNS host and publish the DS record at GoDaddy.
4	Harden email authentication	Move SPF from ~all to -all once all senders are listed, and add DKIM and a DMARC policy (p=quarantine, then p=reject).
5	Limit the REST API	Restrict /wp-json/wp/v2/users to logged-in users with a security plugin or a ModSecurity rule.
6	Keep patching	Keep WordPress core and plugins updated, as the site already does (7.0.4 → 7.1.4).
7	Keep the WAF tuned	Keep ModSecurity with the OWASP Core Rule Set enabled and review its logs.
8	Know your LAN	Scan your own network regularly, keep an inventory of IP/MAC addresses, and investigate any unknown device.

10. Conclusion & Lessons Learned

Week 2 showed how much an attacker can learn before sending a single attack packet. Six harmless, read-only tools revealed the registrar, hosting provider, IP addresses, exact CMS and plugin versions, firewall, caching stack, mail setup and DNS weaknesses of a live website. Zenmap then showed how quickly every device on a local network, with its IP and MAC address, can be mapped.

- **Recon is quiet but powerful:** none of these tools attack, which is why footprinting is so hard to detect.
- **Every tool adds a layer:** no single tool gave the full picture, but six together did.
- **Results change:** the target had patched WordPress and the plugin since the guide was written, so recon must be repeated before each test.
- **Defenders should look first:** running these same tools on your own domain shows exactly what an attacker sees.
- **Adapt without cutting corners:** when my Kali laptop was unavailable, I rebuilt the toolset on another Linux terminal and still completed every task.
- **Stay in scope:** every action here stayed inside the authorized targets and activities.

11. Troubleshooting Log

Issue	Cause	Fix
Kali VM not available	Laptop with the Week 1 lab was under repair	Installed the same six tools on a Debian Linux cloud terminal (apt: whois, whatweb, dnsutils; pip: wafw00f, dnsrecon).
wafw00f / dnsrecon: command not found	pip --user installs to ~/.local/bin, which is not on PATH	export PATH=$HOME/.local/bin:$PATH
Long terminal output cut off in screenshots	whois prints long legal notices	Saved the full output with tee to .txt files and filtered the screenshot to the key fields with grep.
Zenmap needs admin rights	Npcap installs a network driver	Ran the official installer and approved the Windows UAC prompt.
Zenmap didn't appear after launch	The first start is slow and the window opened behind other windows	Opened it from the taskbar thumbnail.
Zenmap's Save Graphic button was hidden	Another window overlapped the maximized Zenmap window	Restored the window to normal size and saved the PDF from there.

12. Evidence Collected

File	Description
evidence/pm1_01_whois.jpg + outputs/01_whois.txt	WHOIS registration record
evidence/pm1_02_whatweb.jpg + outputs/02_whatweb.txt	WhatWeb technology fingerprint
evidence/pm1_03_nslookup.jpg + outputs/03_nslookup.txt	DNS resolution
evidence/pm1_04_curl.jpg + outputs/04_curl_headers.txt	HTTP response headers
evidence/pm1_05_wafw00f.jpg + outputs/05_wafw00f.txt	WAF detection
evidence/pm1_06_dnsrecon.jpg + outputs/06_dnsrecon.txt	DNS enumeration
evidence/pm5_01_install.jpg	Zenmap 7.993 installed (Task 1)
evidence/pm5_02_ipconfig.jpg	IP, subnet, MAC and gateway via nmap --iflist (Task 2)
evidence/pm5_03_pingscan.jpg	Ping scan of 192.168.8.0/24 (Tasks 3–6)
evidence/pm5_04_hosts.jpg	Host details with MAC address
evidence/pm5_05_topology.jpg + evidence/Zenmap_Topology_Risny_WK2.pdf	Topology with legend, saved as PDF (Task 7)

Appendix A — Letter of Authorization (scope summary)

Testing was carried out under the Networkwalks internship Letter of Authorization for cybersecurity testing (Week 2 template). The scope I followed:

Client / target owner	Networkwalks
Authorized tester (intern)	Junaideen Risny Suha
Program / batch	B084 — Networkwalks Cybersecurity Internship
In-scope targets	networkwalks.com and its DNS records; the tester's own LAN
Authorized activities	Footprinting & reconnaissance (whois, whatweb, nslookup, curl, wafw00f, dnsrecon); host discovery with Zenmap / Nmap ping scan
Prohibited	Exploitation, privilege escalation, DoS, brute-force, data access/modification, social engineering, any out-of-scope target

Acknowledged by (Tester): Junaideen Risny Suha, Intern, Batch B084

Signature: _____ Date: 13 November 2026